

RECAP: Fix a prob. space  $(\Omega, \mathcal{F}, \mathbb{P})$

Consider a Time Homogeneous DTMC  $\{X_n : n \in \mathbb{N}\}$  on the state space  $S$  with transition probability matrix  $P$ .  $P_{ij} = \mathbb{P}\{X_{n+1} = j | X_n = i\}$ .

- $n$ -step transition probabilities  $\mathbb{P}\{X_{n+k} = j | X_k = i\}$
- Chapman-Kolmogorov Equation: i)  $P^{(n)} = P^n$ ,  $\forall n \in \mathbb{N}$   
ii)  $\pi(n+1) = \pi(n)P$ ,  $\forall n \in \mathbb{N}$

where  $\pi(n)$  is the unconditional PMF of  $X_n$ .

- finite dimensional distributions of a TH DTMC with TPM  $P$   
→ The initial PMF  $\pi(1)$  and the TPM  $P$  completely specifies all the FDDs.

PLAN: 1) Stopping Time - Definition, Example.

2) Strong Markov Property of a TH DTMC

3) Hitting Times and Recurrence

REFERENCE: A. Kumar, "Discrete Event Stochastic Processes".

Strong Markov Property: Fix a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$

Suppose that  $\{X_n : n \in \mathbb{N}\}$  is a time homogeneous DTMC on the state space  $S$ .

Markov Property: For a TH-DTMC with TPM  $P$ , for any  $n \in \mathbb{N}$

$$\mathbb{P}\{X_{n+1} = j | X_1 = i_1, \dots, X_{n-1} = i_{n-1}, X_n = i\} = \mathbb{P}\{X_{n+1} = j | X_n = i\} = P_{ij}$$

$$\text{i.e. } X_{n+1} \perp \{X_1, \dots, X_{n-1}\} \mid X_n \text{ and } \mathbb{P}\{X_{n+1} = j | X_n = i\} = P_{ij}$$

This is w.r.t a fixed  $n$ . Suppose  $T$  is a random time, i.e.  $T$  itself is a random variable taking values in the index set  $\mathbb{N}$  of the DTMC, i.e.  $T: \Omega \rightarrow \mathbb{N}$ .

Does the Markov Property hold when  $n$  is replaced by  $T$ ?

Remark: [Gallager, "Stochastic Processes Theory for Applications", Chapter 4]

Random variables  $X_1, X_2, X_3, X_4$  on  $(\Omega, \mathcal{F}, \mathbb{P})$  are said to satisfy the Markov property if  $\mathbb{P}\{X_4 | X_1, X_2, X_3\} = \mathbb{P}\{X_4 | X_3\}$  and  $\mathbb{P}\{X_3 | X_1, X_2\} = \mathbb{P}\{X_3 | X_2\}$

But,  $\mathbb{P}\{X_4 = j | X_3 = i\}$  need not be equal to  $\mathbb{P}\{X_3 = j | X_2 = i\}$

However, when we say  $X_1, X_2, X_3, X_4$  forms a Time Homogeneous Markov Chain, denoted  $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$ , it implies not just the Markov property but also that  $\mathbb{P}\{X_4 = j | X_3 = i\} = \mathbb{P}\{X_3 = j | X_2 = i\} = \mathbb{P}\{X_2 = j | X_1 = i\} = p_{ij}$  where  $P$  is the TPM of the TH-DTMC.

So, for a time homogeneous DTMC  $\{X_n, n \in \mathbb{N}\}$  on the state space  $\Omega$  with TPM  $P$ , the question whether the Markov property holds w.r.t a random time  $T$  is not just about whether the random variables are Markovian but also whether the transition probabilities are homogeneous and governed by the entries in the TPM  $P$ .

i.e., the question is:  $\mathbb{P}\{X_{T+k} = j | X_0 = i_0, \dots, X_T = i\} = \mathbb{P}\{X_{T+k} = j | X_T = i\} = (P^k)_{ij}$ ?

YES, if  $T$  is a stopping time.

Example 1: Fix a prob. space  $(\Omega, \mathcal{F}, \mathbb{P})$ . Consider a TH DTMC  $\{X_n: n \in \mathbb{N}\}$  on state space  $S$ . For  $j \in S$ , let  $T_j$  be the random time when the process first visits the state  $j$ , i.e.  $T_j(\omega) = k$  if  $X_0(\omega) \neq j, X_1(\omega) \neq j, \dots, X_{k-1}(\omega) \neq j, X_k(\omega) = j$ . If for some  $\omega \in \Omega$ ,  $\nexists k$  s.t.  $T_j(\omega) = k$ , then  $T_j(\omega) = \infty$ .

Is  $\mathbb{P}(X_{T+k} = j | X_1 = i_1, \dots, X_T = i) = P_{ij}^{(k)}$ ?

Example: Fix a prob. space  $(\Omega, \mathcal{F}, P)$ . Consider a TH DMC  $\{X_n: n \in \mathbb{N}\}$  on the state space  $S$  with TPM  $P$ . Define  $U^{(j)}$  to be time of second visit / first return to state  $j \in S$ . Let  $T = U^{(i)} - 1$ , i.e.,  $T \geq 1$  is one step before the second visit to state  $j$ . Now, for some  $i, k \in S$ .

$$P(X_{T+1} = k \mid X_0 = i_0, \dots, X_T = i) = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{otherwise} \end{cases} \neq p_{i,k} \text{ in general.}$$

$$P\{X_{T+1} = k \mid X_0 = i_0, \dots, X_T = i\} = P\{X_{T+1} = k \mid X_T = i\} \text{ trivially.}$$

$$\text{i.e., } X_{T+1} \perp\!\!\!\perp X_0, \dots, X_{T-1} \mid X_T. \text{ (Markovian)}$$

What does NOT hold is  $P\{X_{T+1} = k \mid X_T = i\} = p_{i,k}$ .

Qn: Why are we interested in random times, and the existence of Markov property w.r.t random times?

Example:  $\{X_n: n \in \mathbb{N}\}$  models a queue. We are interested in the distribution of sojourn time (waiting time) of a random arrival. The arrival instant  $T$  is a random time, and in order to study the sojourn time, we need to look at the future after the random time  $T$  of the queue process.

## Stopping Time

Defn: A random time  $T$  is said to be a **stopping time** for the random process  $\{X_n, n \in \mathbb{N}\}$  on the state space  $S$ , if  $\forall n \in \mathbb{N}$ ,  $\exists$  a function  $f_n: S^n \rightarrow \{0, 1\}$  s.t.  $\forall n \in \mathbb{N}$ ,  $\mathbb{I}\{T \leq n\}(\omega) = f_n(X_1(\omega), \dots, X_n(\omega))$ , i.e., to answer the question whether  $T(\omega) \leq n$  or not, we only need to look at  $X_1(\omega), \dots, X_n(\omega)$ .

Examples: Consider a TH DTMC  $\{X_n, n \in \mathbb{N}\}$  on the state space  $S$ .

Let a sample path be  $X_1 = i_1, X_2 = i_2, \dots$   $i_1, i_2, \dots \in S$ .

1)  $T^{(j)}$  = the time of first visit to state  $j \in S$ . Is this a stopping time?

$$I_{\{T_j \leq n\}} = \begin{cases} 1 & \text{if } j \in \{i_1, i_2, \dots, i_n\} \\ 0 & \text{otherwise} \end{cases}$$

2)  $Z^{(j)}$  = the time of last visit to state  $j \in S$ . Is this a stopping time?

3)  $U^{(j)}$  = time of second visit to  $j \in S$ . Is this a stopping time?

$$I_{\{U^{(j)} \leq n\}} = \begin{cases} 1 & \text{if } \exists n_1, n_2 \leq n \text{ s.t. } i_{n_1} = j \text{ and } i_{n_2} = j, n_1 \neq n_2 \\ 0 & \text{otherwise} \end{cases}$$

4)  $T = U^{(j)} - 1$

Not a stopping time!

$$I_{\{T \leq n\}} = \begin{cases} 1 & \text{if } \exists n_1, n_2 \leq n+1 \text{ s.t. } i_{n_1} = j \text{ \& } i_{n_2} = j, n_1 \neq n_2 \\ 0 & \text{otherwise} \end{cases}$$

5)  $T = m$  ( $m$  deterministic constant) is a stopping time since

$$I_{\{T \leq n\}} = \begin{cases} 1 & \text{if } m \leq n \\ 0 & \text{otherwise} \end{cases}$$

Theorem: Fix a probability space  $(\Omega, \mathcal{F}, P)$ . Let  $\{X_n, n \in \mathbb{N}\}$  be a TH DTMC with state space  $S$  and let  $T$  be a stopping time s.t.  $P\{T < \infty\} = 1$ . Then, the DTMC has the **strong Markov property** w.r.t  $T$ , i.e.,

$$P\{X_{T+s} = j \mid X_1 = i_1, X_2 = i_2, \dots, X_T = i\} = P_{i,j}^{(S)}$$

Proof: 
$$P(X_1 = i_1, \dots, X_T = i, X_{T+s} = j) = \sum_{t=1}^{\infty} P(X_1 = i_1, \dots, X_t = i, X_{T+s} = j, T = t)$$
  

$$= \sum_{t=1}^{\infty} P(X_1 = i_1, \dots, X_t = i) P(X_{t+s} = j \mid X_1 = i_1, \dots, X_t = i) P(T = t \mid X_1 = i_1, \dots, X_t = i, X_{t+s} = j)$$

By Markov property of  $\{X_n, n \in \mathbb{N}\}$  with respect to fixed time  $t$ .

By definition of stopping time  $\{T=t\} \perp X_{t+s}$  given  $\{X_1, \dots, X_t\}$

$$= \sum_{t=1}^{\infty} P(X_1=i_1, \dots, X_t=i) P_{i,j}^{(s)} P(T=t | X_1=i_1, \dots, X_t=i)$$

$$= P_{i,j}^{(s)} \sum_{t=1}^{\infty} P(X_1=i_1, \dots, X_t=i, T=t)$$

$$= P_{i,j}^{(s)} P(X_1=i_1, \dots, X_T=i)$$

$$\text{We have: } P(X_1=i_1, \dots, X_T=i, X_{T+s}=j) = P_{i,j}^{(s)} P(X_1=i_1, \dots, X_T=i)$$

$$P(X_{T+s}=j | X_1=i_1, \dots, X_T=i) = P_{i,j}^{(s)}$$

divide by this on both sides.  
(if positive)

## Hitting and Recurrence Times

Let  $\{X_n, n \in \mathbb{N}\}$  be a TH DTMC on a state space  $S$  with TPM  $P$ .

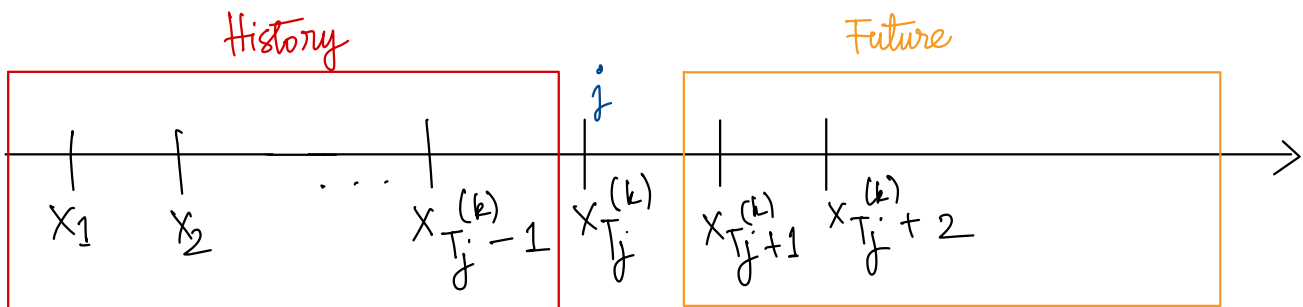
For  $j \in S$ , let  $T_j^{(0)} \triangleq 0$ , and

$$T_j^{(k)} = \min\{n > T_j^{(k-1)} : X_n = j\}, \quad k \in \mathbb{N}$$

For each  $k \in \mathbb{N}$ , the random variable  $T_j^{(k)}$  is the  $k^{\text{th}}$  hitting time of state  $j$ .

For each  $k \in \mathbb{N}$ , suppose  $P\{T_j^{(k)} < \infty\} = 1$ , then the history upto (not including)

$T_j^{(k)}$  is independent of the future unconditionally



Proof: According to Strong Markov Property:

$$(X_1, X_2, \dots, X_{T_j^{(k)}-1}) \perp\!\!\!\perp (X_{T_j^{(k)}+1}^{(1)}, X_{T_j^{(k)}+2}, \dots) \mid X_{T_j^{(k)}}^{(k)}$$

$X_{T_j^{(k)}}^{(k)}$  is a constant random variable (taking the constant value  $j$ )

Thus conditioning on  $X_{T_j^{(k)}}^{(k)}$  is as good as not conditioning at all.

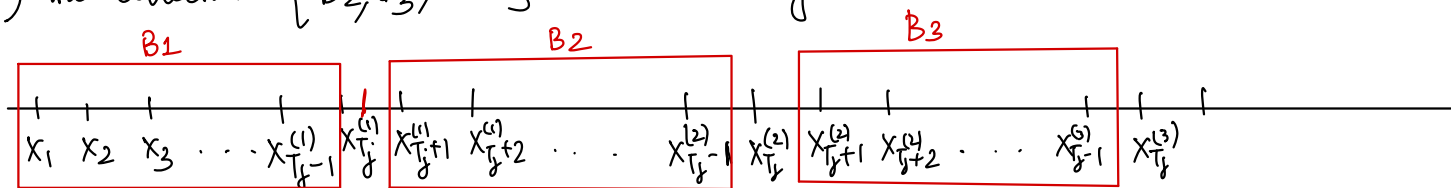
### IID Block Structure

For each  $k \geq 0$ , define the  $k^{\text{th}}$  block as

$$B_k \triangleq \{X_{T_j^{(k-1)}+1}^{(k-1)}, X_{T_j^{(k-1)}+2}, \dots, X_{T_j^{(k)}-1}^{(k)}\}$$

Suppose that  $P(T_j^{(k)} < \infty) = 1 \quad \forall k$  then, the following holds.

- 1) The collection  $\{B_1, B_2, B_3, \dots\}$  are independent
- 2) The collection  $\{B_2, B_3, \dots\}$  are identically distributed



$$(X_1, X_2, \dots, X_{T_j^{(k)}-1}^{(k)}) \perp\!\!\!\perp (X_{T_j^{(k)}+1}^{(k)}, X_{T_j^{(k)}+2}^{(k)}, \dots) \quad \therefore B_k \perp\!\!\!\perp B_{k+1}$$

To show identical distribution of  $B_2, B_3, \dots$ , it suffices to show that

$$X_{T_j^{(1)}+1}^{(1)} \stackrel{d.}{=} X_{T_j^{(2)}+1}^{(2)} \stackrel{d.}{=} \dots$$

$$\text{For any } k \in \mathbb{N}, \quad P\{X_{T_j^{(k)}+1}^{(k)} = i\} = P\{X_{T_j^{(k)}+1}^{(k)} = i \mid X_{T_j^{(k)}}^{(k)} = j\} = P_{ij}.$$

Recurrence Times: The  $k^{\text{th}}$  recurrence time associated with state  $j$  is defined as

$$H_j^{(k)} = T_j^{(k+1)} - T_j^{(k)} \quad k \in \mathbb{N}.$$

$H_j^{(k)}$  is the (random) length of the block  $B_{k+1}$

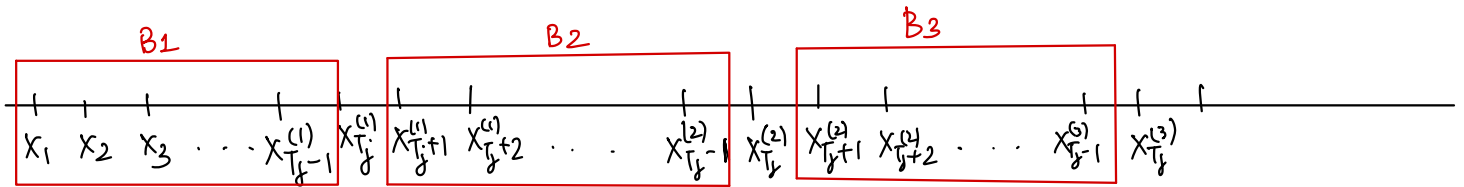
## i.i.d Property of recurrence times

Suppose that  $P(T_j^{(k)} < \infty) = 1 \quad \forall k \in \mathbb{N}$ . Then

$\{H_j^{(k)}\}_{k=1}^{\infty}$  is a sequence of i.i.d random

### Variables:

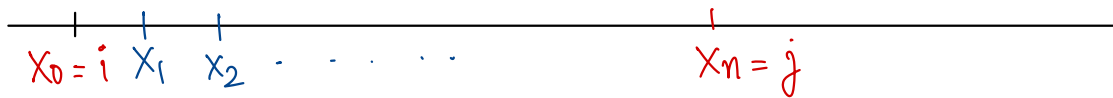
The result follows from the i.i.d nature of the blocks  $B_2, B_3, \dots$



First Passage Time Distribution: Fix a probability space  $(\Omega, \mathcal{F}, P)$

Given that  $\{X_n, n \geq 0\}$  is a DTMC on the state space  $S$ .

For  $i, j \in S$ ,  $n \geq 1$ , define  $f_{i,j}^{(n)} = P(X_1 \neq j, X_2 \neq j, \dots, X_{n-1} \neq j, X_n = j | X_0 = i)$   
 i.e.,  $f_{i,j}^{(n)}$  is the probability that the DTMC hits  $j$  for the first time at the  $n^{\text{th}}$  step given that it starts off in state  $i$ .



If  $j = i$ , then  $f_{i,i}^{(n)}$  is the probability of first return to state  $j$  at the  $n^{\text{th}}$  step.

$$f_{i,j} = P(\exists n \geq 1 \text{ s.t. } X_n = j | X_0 = i) = P(T_j^{(1)} < \infty | X_0 = i)$$

probability of eventually hitting state  $j$  if the process starts off in state  $i$ .

$$= P(\bigcup_{n=1}^{\infty} (X_k \neq j, 1 \leq k < n, X_n = j | X_0 = i))$$

$$= \sum_{n=1}^{\infty} f_{i,j}^{(n)}$$

$f_{i,j}^{(n)}$  only depends on TPM

In general  $f_{i,j} \leq 1$

For  $i, j \in S$ ,  $f_{i,j}^{(n)}$  is called the **first passage time distribution** for hitting  $j$  starting from state  $i$ . i.e.,  $f_{i,j}^{(n)}$  is a probability mass fn on the positive integers. This distribution can, in general, be defective (i.e.  $\sum f_{i,j}^{(n)} < 1$ ).



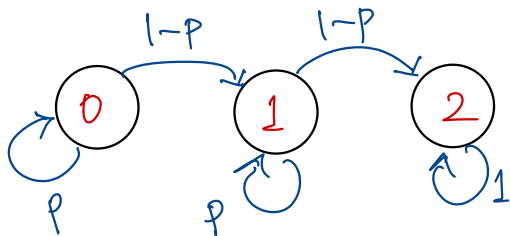
If  $f_{ij} < 1$ , then  $1 - f_{ij} > 0$  is the prob. that the DTMC never visits  $j$  starting from state  $i$ .

When  $f_{jj} = 1$ , the mean time to return, or the **mean recurrence time** is defined to be

$$V_{jj} = \sum_{n=1}^{\infty} n f_{jj}^{(n)}$$

Define  $f_{jj}^{(0)} = 1$  and for  $i \neq j$   $f_{ij}^{(0)} = 0$ .

Examples:

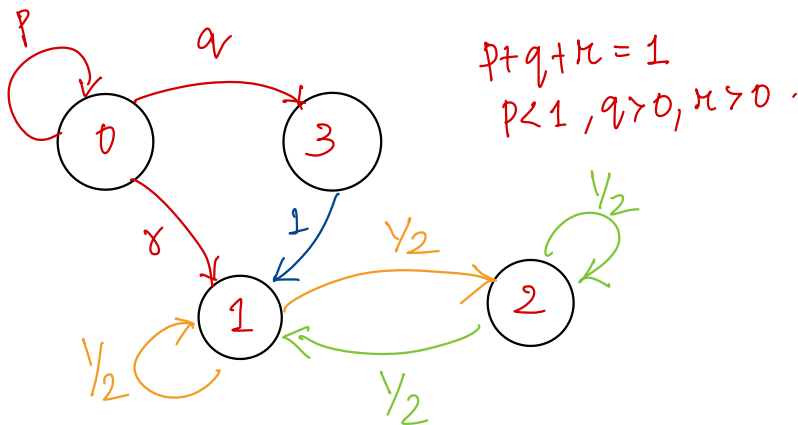


$$\begin{array}{l|l} f_{00}^{(1)} = & f_{00} = \\ f_{00}^{(n)} = & n \geq 2 \end{array}$$

$$\text{For } n \geq 1 \quad \begin{array}{l} f_{01}^{(n)} \\ f_{02}^{(n)} \end{array}$$

$$\begin{array}{l} f_{11} \\ f_{22} \end{array}$$

Ex: 2



$$\begin{array}{l|l} f_{01}^{(1)} & \\ f_{01}^{(k)} & k \geq 2 \\ f_{01} = & \end{array}$$

Why care about hitting times and recurrence?

- Applications:
- 1) Stock Market Analysis : How long till a stock reaches a target price?
  - 2) Analysis of Randomized Algorithms for Boolean-SAT problems with application to Electronic Design Automation (Automatic Test Pattern Gen)